FUNDAMENTALS TO FULL-STACK AI ENGINEER

Complete topic-by-topic learning syllabus • Beginner → AI/ML → LLMs → Full-stack → Production

FORMAT: Headings and topic names only. No long explanations. Areas 05–09 preserve the topic titles from the photos and the previously shared topic list.

AREA 01 — COMPUTING & PROGRAMMING FUNDAMENTALS

Computer Fundamentals

- How computers work
- CPU, RAM, storage and GPU
- Binary and hexadecimal
- Operating systems
- Files and directories
- Environment variables

Programming with Python

- Variables and data types
- Operators and expressions
- Conditionals and loops
- Functions and scope
- Lists, tuples, sets and dictionaries
- Strings and comprehensions
- OOP: classes, objects, inheritance and composition
- Modules, packages and imports
- Exceptions and error handling
- Type hints and dataclasses
- Iterators and generators
- Decorators and context managers

Developer Tools

- Linux terminal and shell
- Git and GitHub
- Virtual environments and pip
- Debugging
- Code formatting and linting
- Testing with pytest
- Jupyter Notebook
- Documentation and README files

Core Computer Science

- Data structures
- Arrays and strings
- Linked lists, stacks and queues
- Hash tables
- Trees, heaps and graphs
- Searching and sorting
- Recursion and backtracking
- Time and space complexity
- Processes and threads
- Concurrency and parallelism
- Networking fundamentals

AREA 02 — MATHEMATICS, DATA & MACHINE LEARNING

Mathematics for AI

- Linear algebra
- Vectors and matrices
- Dot product and matrix multiplication
- Norms, projections and eigenvalues
- SVD and matrix calculus
- Differential and integral calculus
- Derivatives and chain rule
- Partial derivatives and gradients
- Jacobians and Hessians
- Probability and random variables
- Probability distributions
- Conditional probability and Bayes' theorem
- Expectation and variance
- Descriptive and inferential statistics
- Sampling and confidence intervals
- Hypothesis testing
- Optimization and gradient descent
- Entropy and cross-entropy
- KL divergence and mutual information

Data Science & Data Processing

- NumPy
- Pandas
- Data cleaning
- Missing values and outliers
- Exploratory data analysis
- Data visualization with Matplotlib and Seaborn
- Feature engineering
- Encoding and scaling
- Data pipelines
- SQL fundamentals

Machine Learning

- ML workflow
- Supervised and unsupervised learning
- Linear regression
- Logistic regression
- k-Nearest Neighbors
- Naive Bayes
- Decision trees
- Random forests
- Gradient boosting
- XGBoost, LightGBM and CatBoost
- Support Vector Machines
- k-means clustering
- Hierarchical clustering
- DBSCAN
- Principal Component Analysis
- Anomaly detection
- Recommendation systems
- Feature selection
- Regularization
- Overfitting and underfitting
- Bias-variance trade-off

ML Evaluation

- Train, validation and test split
- Cross-validation
- Confusion matrix
- Accuracy, precision and recall
- F1-score
- ROC-AUC and PR-AUC
- Regression metrics: MAE, MSE and RMSE
- Class imbalance
- Data leakage
- Model interpretability and SHAP

AREA 03 — DEEP LEARNING & NEURAL NETWORKS

Neural Network Fundamentals

- Perceptron
- Neurons, weights and biases
- Activation functions
- Forward propagation
- Loss functions
- Backpropagation
- Computational graphs
- Gradient descent and optimizers
- Learning-rate schedules
- Batch size and epochs
- Weight initialization
- Normalization and regularization
- Dropout and batch normalization

Deep Learning Architectures

- Multilayer Perceptrons (MLP)
- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- LSTM and GRU
- Autoencoders
- Variational Autoencoders
- Generative Adversarial Networks (GANs)
- Attention mechanism
- Encoder-decoder architecture
- Transformers

Deep Learning Frameworks

- PyTorch tensors
- Autograd
- Datasets and DataLoaders
- Building models with nn.Module
- Training and validation loops
- GPU training and CUDA basics
- Checkpointing and experiment tracking
- Model inference and export

AREA 04 — LLMs — LARGE LANGUAGE MODELS

NLP Foundations

- Text preprocessing
- Tokenization
- Word and subword tokens
- Vocabulary and embeddings
- One-hot encoding
- Word2Vec and GloVe
- Language modeling
- N-grams and perplexity

Transformer Architecture

- Self-attention
- Query, Key and Value
- Scaled dot-product attention
- Multi-head attention
- Positional encoding
- Encoder and decoder blocks
- Feed-forward networks
- Residual connections and LayerNorm
- Causal masking
- KV cache

LLM Training & Adaptation

- Pretraining
- Next-token prediction
- Dataset preparation and filtering
- Training objectives
- Instruction tuning
- Supervised fine-tuning (SFT)
- Preference datasets
- RLHF
- DPO
- LoRA and QLoRA
- Parameter-efficient fine-tuning
- Quantization
- Distillation

LLM Usage & Inference

- Context windows
- Prompt design
- System, developer and user messages
- Temperature and sampling
- Top-k and top-p
- Structured outputs
- Function calling and tool use
- Open-source vs hosted models
- Hugging Face Transformers
- Ollama
- vLLM

RAG — Retrieval-Augmented Generation

- Document ingestion
- Text extraction and parsing
- Chunking strategies
- Embeddings
- Vector databases
- Similarity search
- Metadata filtering
- Hybrid search
- Reranking
- Context assembly
- Grounded generation
- Citations and source attribution
- RAG evaluation

AI Agents & Advanced LLM Apps

- Agent architectures
- Tool calling
- Planning and reasoning loops
- Memory systems
- State and context management
- Workflow orchestration
- LangChain
- LangGraph
- Multi-agent systems
- MCP: Model Context Protocol
- Human-in-the-loop workflows
- Multimodal LLMs
- Vision-language models
- Speech and audio models

AREA 05 — EVALS — AI QUALITY & OBSERVABILITY

Evaluation Foundations

- Golden Datasets
- Error Analysis
- LLM as a Judge
- Pairwise Comparison
- Rubric Scoring
- Faithfulness
- Answer Relevance
- Context Precision
- Tool Call Accuracy
- Trajectory Evals

Evaluation Operations

- Regression Evals in CI
- A/B Testing Prompts
- Human Labeling
- Synthetic Test Data
- Tracing
- Langfuse
- OpenTelemetry for LLMs
- P95 Latency
- Cost per Request
- User Feedback Loops

AREA 06 — AI SYSTEM DESIGN

Model & Request Architecture

- Model Routing
- Model Fallbacks
- Semantic Caching
- Prefix Caching
- Token Budgets
- Streaming Responses
- Time to First Token
- Throughput vs Latency

Inference & Performance

- Inference Serving with vLLM
- Continuous Batching
- PagedAttention
- Speculative Decoding
- GPU vs API Cost
- Batch Inference

Scalable AI Architecture

- Async Job Pipelines
- Provider Rate Limits
- Multi-tenant Design
- RAG vs Fine-tuning
- Context Window Management
- Cost Optimization

AREA 07 — AI SECURITY

LLM Threats & Defenses

- Prompt Injection
- Indirect Prompt Injection
- Jailbreaks
- OWASP Top 10 for LLMs
- Input Guardrails
- Output Validation
- PII Redaction
- Data Leakage

Agent & Infrastructure Security

- Least-privilege Tools
- Tool Sandboxing
- Secrets Management
- API Key Rotation
- Authorization for Agents
- Human Approval for Actions
- Content Moderation
- Data Poisoning
- Access Control in RAG
- MCP Server Trust
- Audit Logging
- Red Teaming

AREA 08 — DEPLOYMENT & MLOPS / LLMOPS

Containers & Infrastructure

- Docker
- Dockerfiles and images
- Docker Compose
- Kubernetes basics
- GPU instances and drivers
- Cloud fundamentals
- Infrastructure as Code
- Serverless functions

CI/CD & Release Engineering

- CI/CD pipelines
- GitHub Actions
- Automated testing
- Eval gates in CI
- Environment configs
- Model versioning
- Prompt versioning
- Canary releases
- Feature flags
- Rollbacks

Production Operations

- Ollama for local models
- Autoscaling
- Load balancing
- Health checks
- Logging and monitoring
- Alerts and dashboards
- Experiment tracking with MLflow
- Model registry
- Data and model drift
- Scheduled retraining
- Secrets and configuration management

AREA 09 — BACKEND ENGINEERING FOR AI

API Development

- HTTP and HTTPS
- REST APIs
- HTTP methods and status codes
- JSON
- FastAPI
- Request and response models
- Input validation with Pydantic
- Async and await
- Streaming with SSE
- WebSockets
- API documentation

Databases & Storage

- SQL and relational databases
- PostgreSQL
- Database schema design
- Joins and indexes
- Transactions and ACID
- NoSQL fundamentals
- MongoDB
- Vector databases
- Redis caching
- File storage and object storage

Backend Reliability

- Authentication with JWT
- OAuth 2.0
- Authorization and RBAC
- Rate limiting
- Message queues
- Background jobs
- Retries with backoff
- Idempotency keys
- Pagination
- Logging and monitoring
- Unit and integration testing

AREA 10 — FRONTEND & FULL-STACK AI APPLICATIONS

Web Foundations

- HTML5
- CSS3
- Responsive design
- JavaScript fundamentals
- DOM and browser APIs
- ES6+ modules
- Promises and async/await
- TypeScript

Frontend Frameworks

- React
- Components and props
- State and hooks
- Forms and validation
- Routing
- API integration
- Error and loading states
- Next.js fundamentals
- Frontend testing

AI Product Interface

- Chat interfaces
- Streaming token display
- Conversation history
- File and document uploads
- Source citations UI
- Tool activity and status UI
- Human approval screens
- Feedback controls
- Authentication UI
- Usage and billing dashboards

Full-Stack Integration

- Frontend-backend integration
- API contracts
- Authentication flow
- Database integration
- RAG application integration
- Agent workflow integration
- File processing pipeline
- Deployment of frontend and backend
- End-to-end testing

AREA 11 — CAPSTONE — PRODUCTION-READY FULL-STACK AI ENGINEER

Portfolio Projects

- AI-powered document assistant with RAG
- Research assistant with citations
- Multi-tool AI agent
- LLM evaluation dashboard
- Secure multi-tenant AI SaaS

Final System Skills

- System architecture and design
- Latency and cost optimization
- Security and privacy review
- Observability and tracing
- Load and reliability testing
- CI/CD and production deployment
- Technical documentation
- GitHub portfolio and project demos